Hisham Almalki

Computer Science Graduate | Data Science & Data Analysis

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SUMMARY

Computer Science graduate from Umm Al-Qura University. During cooperative training at the Deanship of Information Technology, led a trainee team developing a mobile app for lab computer inspection and task management, and built a Power BI report on student withdrawal data. Skilled in Python, SQL, and Excel for data analysis, with projects in machine learning, computer vision, and LLM-powered web applications.

EDUCATION

Bachelor of Computer Science, *Umm Al-Qura University, College of Computers* 08/2022 – 06/2026
GPA: 3.48 / 4.00 | Second Class Honors Makkah, Saudi Arabia

EXPERIENCE

Cooperative Training, *Deanship of Information Technology, Umm Al-Qura University* 03/2025 – 08/2025

- Led a 4-member trainee team to prototype a Flutter mobile app replacing Google Forms for lab device inspections, addressing duplicate records and limited progress tracking.
- Developed Firebase database components and integrated them with the app interface, covering device records, lab progress, and assigned tasks.
- Co-authored the app's Software Requirements Specification (SRS) and presented the prototype to Deanship leadership.
- Built a Power BI report on student withdrawal data at the supervisor's request, identifying the colleges and nationalities with the highest number of withdrawals.
- Inventoried 200+ computer devices across university labs during a rotation with the Labs Department.

PROJECTS

Faten, *AI Academic Assistant (Graduation Project)*

- Part of a team that built a full-stack academic platform, piloted at Umm Al-Qura University's College of Computers, featuring an LLM chat agent, GPA analytics, AI study planning, and indoor navigation across 89+ rooms.
- Built function-calling tools for the platform's OpenAI-powered chat agent, including academic performance analysis that identifies students' strengths and weaknesses from their records, exam date lookup, and course information with grading breakdowns.
- Built Firebase components for storing and retrieving student academic data, used by the agent to personalize its responses.

Pulmonary Edema Detector, *Chest X-ray Classification*

- Part of a three-member team that built a Streamlit app classifying chest X-rays as normal or pulmonary edema, with Grad-CAM heatmaps explaining each prediction.
- Collected and assembled 10,803 chest X-ray images from multiple public sources, including Kaggle and Radiopaedia.
- Fine-tuned DenseNet121 and EfficientNet-B0 in PyTorch with data augmentation; the team's four CNNs reached 0.974–0.979 ROC-AUC on a held-out test set.
- Built the soft-voting ensemble combining all four models.

Telecom Customer Churn Analysis, *Data Analysis & Machine Learning*

- Cleaned 7,043 customer records in Excel and Power Query, and built a three-page Power BI dashboard with DAX sizing churn losses at $139K in monthly revenue (26.5% churn rate).
- Identified month-to-month fiber-optic customers (54.6% churn) as the segment carrying ~60% of lost lifetime value, using cross-segment analysis to separate real churn drivers from confounded ones.
- Proposed three prioritized retention actions, including bundling tech support into fiber plans (41.6% churn without support vs. 15.2% with).
- Trained a logistic regression model in Python reaching 0.849 ROC-AUC and 78% recall on churners, after removing target-leakage columns; its coefficients confirmed the drivers found in the dashboard.

SKILLS

Languages: Python, SQL
Data & BI: Excel, Power Query, Power BI, Pandas, NumPy, Matplotlib, Data Cleaning, Data Visualization
Machine Learning & AI: scikit-learn, PyTorch, CNNs, Computer Vision, Grad-CAM, LLM APIs
Tools: Firebase, Git, GitHub, Jupyter
Soft: Leadership, Problem Solving, Adaptability, Technical Presentations, Team Collaboration

CERTIFICATIONS

NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL), *NVIDIA* 10/2026

COURSES

Artificial Intelligence Program, *KAUST Academy*

Data Science and Machine Learning with Python, *Tuwaiq Academy*

Data Analytics Bootcamp, *ATHAR Academy*

Fundamentals of Artificial Intelligence, *SDAIA*

LANGUAGES

Arabic **English**